

Abstract

Addiction is a multidimensional phenomenon influenced by a combination of biological, psychological, and social factors. One of humanity's most important questions is: what happens in the body in response to addiction and the substances derived from drug use? Is it difficult to quit them? Can science advance to a point where we gain a deep understanding of this process, enabling us to help individuals overcome this habit more easily? Or even, could we find a substitute—such as a medication—with minimal side effects to replace it?

Introduction

The value of an experience or action is determined by the reward that follows it.

Addiction, at first glance, must hold a greater value in the brain than other activities. It is believed that certain long-lasting changes in synaptic plasticity are the underlying cause of addiction; these persistent changes lead to disruptions in the brain's reward system. Therefore, in addition to the brain, where decisions are made, there must also be a reward mechanism involved.

On the other hand, there is a psychological theory known as the opponent-process theory, proposed by Richard Solomon, which is considered one of the key frameworks for understanding individuals struggling with addiction.

In the following sections, we will explore the biological and psychological aspects of addiction, followed by an examination of the proposed model and the simulation results presented in the paper.

Cognitive and Biological Aspects

The paper begins with the statement:

“The value of an experience or action is determined by the reward received afterward.”

In our body, there is a mechanism responsible for assigning value to actions. The value associated with nicotine addiction exceeds that of other rewards, leading to the unconscious execution of nicotine intake behaviors in various forms. Biologically, nicotine also induces changes in synaptic plasticity. Over time, the value-assignment mechanism itself undergoes changes, which can be explained by the opponent-process theory (a concept that will be discussed later).

Dopamine is a neurotransmitter that plays a key role in the sensation of reward and reinforcement of specific behaviors, such as drug use. Nicotine influences dopaminergic signaling in the Ventral Tegmental Area (VTA), a region of the brain that plays a critical role in the reward system and is involved in generating feelings of pleasure and learning from rewards (like nicotine). Glutamatergic processes also contribute to the automatic selection of behaviors that lead to nicotine consumption.

Moreover, brain synapses exhibit plasticity. In addicted individuals, changes occur in how neurons communicate with one another. These changes cause a persistent tendency toward engaging in specific behaviors (such as smoking). This synaptic plasticity is considered a major factor in addiction, as it leads to destructive behaviors being performed automatically and uncontrollably.

The model is also based on the following hypothesis:

“Nicotine addiction is a transition from impulsive behavior to compulsive behavior.”

That is, as a result of reinforcement learning, nicotine addiction gradually shifts from impulsive behaviors (performed randomly) to compulsive behaviors (performed unconsciously).

The opponent-process theory is a psychological and neurobiological theory used to explain emotional and motivational experiences in response to stimuli such as drugs, emotions, or intense experiences.

The opponent-process theory states that every intense emotional experience (such as pleasure or discomfort) is typically followed by an opposing emotional state. For example, nicotine initially produces a pleasant feeling (euphoria), but over time, an opposing process is activated (such as anxiety or intense craving), aiming to restore the body's emotional balance.

As a result, individuals continue using nicotine not for the initial pleasure, but to escape the negative state caused by the opponent process.

Proposed Model

Two main approaches to explaining addiction — the opponent-process theory and reward-based learning — are simultaneously considered in developing a computational model of nicotine addiction.

Reward-based learning is grounded in the principle that any action followed by a reward is more likely to be repeated in the future. This artificial reward causes the brain to learn that *smoking* is a highly valuable behavior, even if it leads to negative consequences. On the other hand, individuals continue using nicotine not for the initial pleasure, but to relieve the negative emotional state caused by the opponent process.

The modeling is based on two modules:

- Action selection module, which is based on the functioning of the cortex–basal ganglia–thalamus loop.
- DA signaling module, which has two components: An evaluation component that depends on the presence of nicotine and a value-assignment component related to the reward received from the action and the prediction error, which is defined as the difference between the actual reward and the expected reward from performing the action.

Each of these modules is governed by differential equations, which will be presented in the following sections.

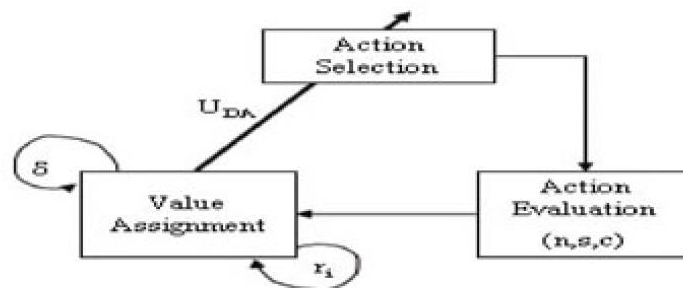


Figure 1 (Model's Module)

Overall, this paper uses difference equations and nonlinear dynamics to model neural processes and the effects of nicotine on the brain. These equations and dynamic systems are specifically designed to simulate how human behavior changes in response to nicotine exposure and addiction.

Action Selection

The Action Selection Module forms the core of the model, simulating how a behavior (such as smoking or not smoking) is chosen. This module consists of two main loops: the Premotor Loop and the Motor Loop, which together model the Cortex–Basal Ganglia–Thalamus (C-BG-TH) circuits.

The Premotor Loop is responsible for the planning and preparation of an action — it decides what action should be taken and how to prepare for it, before the action is actually executed. In contrast, the Motor Loop is directly responsible for the execution of that planned action, producing the actual physical behavior.

Premotor Loop State Variables	Description
$p_{pm}(k)/p_m(k)$	Activity in the Premotor/Motor pathway.
$m_{pm}(k)/m_m(k)$	Activity of the Premotor/Motor circuits.
$r_{pm}(k)/r_m(k)$	Activity related to striatal influence.
$n_{pm}(k)/n_m(k)$	Neuronal activity in the Premotor/Motor area.
$d_{pm}(k)/d_m(k)$	Activity related to the diffuse effect of the subthalamic nucleus in the Premotor/Motor area.

Action Selection Equations

Premotor

$$p_{pm}(k+1) = f(\lambda p_{pm}(k) + m_{pm}(k) + W_{c_{pm}} I(k))$$

$$m_{pm}(k+1) = f(p_{pm}(k) - d_{pm}(k))$$

$$r_{pm}(k+1) = W_{r_{pm}} f(p_{pm}(k))$$

$$n_{pm}(k+1) = f(p_{pm}(k))$$

$$d_{pm}(k+1) = f(W_{d_{pm}} n_{pm}(k) - r_{pm}(k))$$

Motor

$$p_m(k+1) = f(\lambda p_m(k) + m_m(k) + \beta p_{pm} + noise)$$

$$m_m(k+1) = f(p_m(k) - d_m(k))$$

$$r_m(k+1) = W_{r_m} f(p_m(k))$$

$$n_m(k+1) = f(p_m(k))$$

$$d_m(k+1) = f(W_{d_m} n_m(k) - r_m(k))$$

f is activation function.

$$f(x) = \frac{1 + \tanh(a(x - 0.45))}{2}$$

The equations of the Action Selection Module, which include the Premotor and Motor loops, model the neural dynamics involved in behavioral decision-making. These equations use nonlinear activation functions (f) to simulate how sensory inputs ($I(k)$) are processed, how internal brain activities (such as p_{pm} , p_m , m_{pm} , m_m) are regulated, and how various brain structures — such as the subthalamic nucleus (via W_d) and the striatum (via W_r) — exert their influence.

The ultimate goal of these equations is to select an action (e.g., to smoke or not to smoke), with the decision determined by the output activity of p_m .

The model parameters are as follows:

λ : This parameter (with a value of 0.5) controls the self-feedback of activities p_{pm} and p_m within the Premotor and Motor loops, respectively.

β : This parameter (with a value of 0.03) determines the influence and connection of p_{pm} activity (from the Premotor loop) on p_m activity in the Motor loop

a : This parameter (with a value of 3) adjusts the slope of the activation function f , which is used to introduce nonlinearity into the system.

DA Signaling Module Equations

The Dopamine Signaling Module (DA Signaling Module) in the proposed model is a crucial component responsible for processing reward signals and nicotine-related cues. This module, which is activated in the presence of nicotine, consists of two main components: Action Evaluation and Value Assignment.

The Action Evaluation component operates based on the presence of nicotine, while the Value Assignment component calculates the rewards assigned to actions and the prediction error. This module guides the Action Selection (A-S) loop by representing the rewarding value of previously performed actions.

Updates the phasic dopamine activity

$$u_{DA}(k+1) = u_{DA}(k) + \mu_{DA}(-u_{DA}(k) + s_{DA}(ri, Ni))$$

$$s_{DA} = \frac{1 + \tanh(Ni \times ri - \theta_{DA})}{2}$$

Value signal representing the system's evaluation of the current performed action

$$V(k) = (W_v + base)(I(k))$$

Reward prediction error, which is the discrepancy between the immediate reward

$$\delta(k) = ri(k) + \mu V(K+1) - V(k)$$

N-S-C Dynamics

The N-S-C dynamics is an internal subsystem within the model, consisting of three state variables: $n(k)$, $s(k)$, and $c(k)$. This system forms a set of nonlinear difference equations that model the changes in these variables over time. These variables are automatically updated in response to both the external nicotine level and the complex interactions among themselves. The primary goal of this component is to simulate the fundamental neural processes involved in the absorption and processing of nicotine in the brain.

The N-S-C dynamics is directly connected to the dopamine signaling (DA) module, as its output serves as a critical input for that module. Specifically, the variable N_i (nicotine absorption), which is used to compute the dopamine activation signal (S_{DA}), is derived from the product of n and s (when nicotine administration is stopped). This connection ensures that the presence and processing of nicotine in the neural system directly influence dopamine activity, and thus impact reward processing and reinforcement learning. In this way, the N-S-C Dynamics provides the physiological foundation for nicotine's effect on the brain's reward mechanisms and links it to the DA module.

$$\begin{aligned} n(k+1) &= n(k) + \frac{\mu}{\tau_n} [-\beta_n(k)c(k) + \alpha_n(k)(1 - n(k)c(k))] \\ s(k+1) &= s(k) + \frac{\mu}{\tau_s} [-\beta_s s(k) + \alpha_s(k)(1 - s(k))] \\ c(k+1) &= c(k) + \frac{\mu}{\tau_c} [-\beta_c c(k) + \alpha_c(k)(1 - c(k))] \end{aligned}$$

Activation functions:

$$\begin{aligned} \alpha_n(k) &= 0.5(1 + \tanh(\text{nicotine} - \theta_n)) \\ \alpha_s(k) &= 0.5(1 + \tanh(n(k) - \theta_s)) \\ \alpha_c(k) &= 0.5(1 + \tanh(s(k) - \theta_c)) \\ \beta_n(k) &= 0.5(1 + \tanh(c(k) - \theta_n)) \end{aligned}$$

We present a table of the parameters and their initial values below.

Model Parameters and Variables

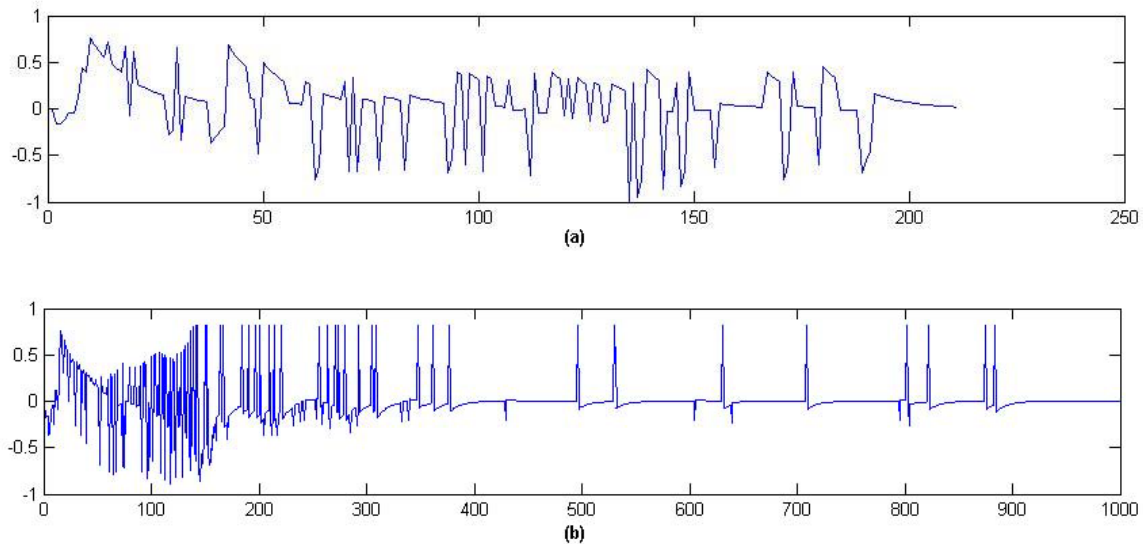
In the table below, we review the parameter values to ensure they are available during implementation.

Parameter / Variable	Explanation	Value
λ	Stability coefficient in premotor/motor loop dynamics	0.5
β	Weight of premotor signal into motor loop	0.03
a	Sharpness of the sigmoid activation function	3
μ_{DA}	Learning rate in dopamine (DA) subsystem	0.1
η_c	Learning rate for W_{cpm} (cortex $\rightarrow$ premotor weights)	0.1
η_v	Learning rate for W_v (stimulus value weights)	0.1
η_r	Learning rate for W_{rpm} (ventral $\rightarrow$ dorsal striatum)	0.1
θ_{DA}	Threshold for tonic dopamine (DA tonic)	0.01
θ	Threshold for premotor signal pm	10% of pm
θ_{wDA}	Threshold for DA in W_{rpm} learning	10% of U_{DA}
$n, s, c (0)$	Initial nicotine-related signals	0.1 each
$\theta_n, \theta_s, \theta_c$	Activation thresholds for n-s-c system	0.6, 0.7, 0.7 respectively
β_s, β_c	Decay factors in s and c equations	0.4
μ	Update rate for n-s-c system	0.1
τ_n, τ_s, τ_c	Time constants for n, s, and c	0.25, 1, 2 respectively
$r_i(0)$	Initial reward value Each time "smoke" is selected Multiplied by 2 until it reaches 1	0.01
Action encoding	"Smoke", "Not smoke", "Undecided"	[1 0], [0 1], [1 1]
W_v, W_c	Initial value weights	Small positive random values
W_{rpm}	Initial ventral to dorsal striatum weights	Diagonal matrix of ones
$W_{dpm/m}$	STN diffusive effect matrices (fixed) and	Matrix with 0.5 values
W_{rm}	Feedback effect from motor to decision (fixed)	Matrix with 0.5 values

Table 1

Result

To evaluate the performance of the proposed nicotine addiction model, the system's response to nicotine uptake was simulated. The initial reward value was set very low and increased with each selection of the "smoking" action until it reached 1. The simulation stops when the smoking action is selected 20 times in a row, which is considered an indication of addiction. In 20 out of 50 runs, addiction was established. The average number of trials required to reach addiction was 346, with a standard deviation of 265.7. These results indicate that the model is capable of representing the process of addiction development.

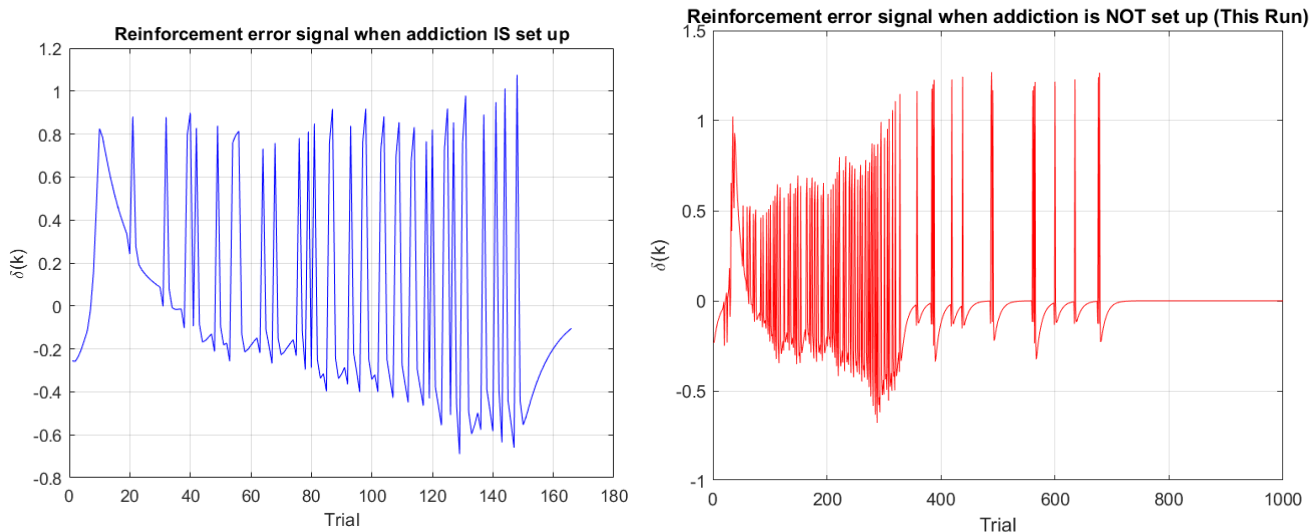


Reinforcement error signal δ when addiction is a) set up b) not set up

Simulation

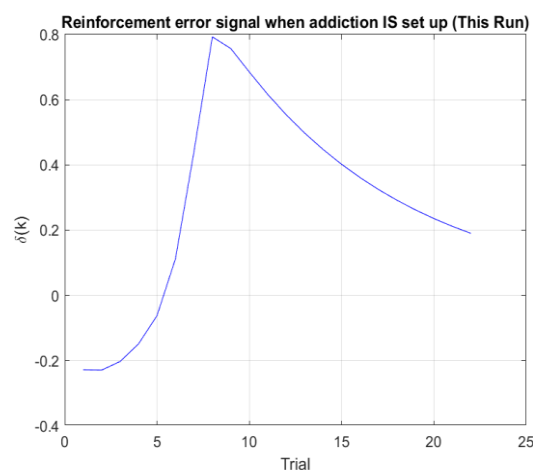
The simulation of this paper was implemented by myself due to the unavailability of the original code. The implementation file is included in the uploaded materials. Modeling brain processes such as addiction requires repetition because of their complexity, nonlinearity, and stochastic nature, in order to obtain reliable and generalizable results. This is why each run of the code may yield different outcomes. Additionally, criteria such as 20 consecutive smoking actions and the use of variance-based metrics are employed precisely because of this relative randomness.

The simulation results are as follows:(In the next slide, we will analyze it.)



Smoker

The initial negative value of δ may indicate that the received reward was less than expected, or that the system's initial expectations are still being adjusted. The rapid increase of δ toward positive values, reaching a peak, represents a significant positive reward prediction error—likely due to the selection of the "smoking" action and receiving a reward that exceeded the initial expectation (since r_i starts at 0.01 and doubles with each selection). On the other hand, the subsequent decrease in δ implies that the agent is gradually learning to correctly predict the value of the "smoking" action, and the difference between expected and received reward is diminishing. In other words, the agent has "learned" that smoking provides a high reward and is no longer "surprised." This decrease and convergence of δ toward zero signals the stabilization of addictive behavior and the transition from random choice to compulsive behavior, as the system consistently and confidently selects the rewarding action.



Non-Smoker

The initial peak and strong fluctuations indicate the system's early attempts to learn and adjust its reward expectations. Large reward surprises (both positive and negative) occur during this phase, as the system is still exploring the relationship between actions and rewards. The failure of δ to converge toward zero suggests that the model was unable to stabilize a consistent, "addictive" behavior (i.e., selecting the smoking action 20 times in a row). The system continues to experience reward prediction errors and cannot consistently predict the value of a specific action (like smoking) with high accuracy. These persistent fluctuations indicate that the system remains in a state of exploration or uncertainty and has not reached a level of "compulsive learning" that would lead to addiction.

Conclusion

This paper examines a model based on real-world theories—namely, the opponent-process theory and reinforcement learning—to explain the valuation of nicotine intake. It mathematically models (using differential equations) the biological and cognitive processes that occur in the body, leading an individual to repeatedly seek out smoking and reintroduce nicotine into the system. Unlike previous models such as the winner-take-all system, this model employs a dynamic system that incorporates previous actions and their evaluations into the decision-making process.

Another goal is to support the idea that addiction develops as a form of goal-directed behavior, and therefore the interaction of cortico-striato-thalamic action-selection loops plays a crucial role in the development of addiction.